Eleanna Vasquez Cerda

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EDUCATION

Ph.D. Pennsylvania State University 2021- Present
Department: Plant Biology Expected defense June 2026
B.A. Mount Holyoke College - South Hadley, MA 2020
Majors: Environmental Studies and History

SKILLS

Software/Platforms: R, Python in Jupyter Notebook, Nextflow, Linux, Bash, SCP, High-performance computing (HPC), SciPy, scikit-learn, Pandas, NumPy, Git, Github
Technical skills: Population genetic statistics (Fst/HFst, SFS, π, Tajima's D, LD, IBD, etc.), parallel computing, supervised (random forest) & unsupervised (PCA) machine learning, statistical tests, predictive modeling (regressions), ordination, Mantel tests, cross-validation
Bioinformatics: NGS data (FASTQ, BAM, VCF), sequence alignment (bwa & samtools), variant calling and annotation (GATK & SNPeff), imputation (Beagle), MALT- MEGAN pipeline, GWAS, BLAST, genomic prediction
Languages: English, Spanish (Native)

RELEVANT PROFESSIONAL EXPERIENCE

Penn. State University, University Park, State College, PA 04/21-Present
Plant Biology Ph.D. Candidate, Thesis Advisor: Jesse Lasky, Ph.D.

- Conducted population genetic analyses to understand how genetic diversity is maintained and distributed across space, and to assess the relative effect of environmental factors contributing to population structure.
- Applied quantitative methods like predictive modeling to analyze large-scale genomic data as well as spatial and climatic data using R and Python in HPC environments.
- Designed supervised and unsupervised machine learning models to make climate projections and assess genetic similarity.
- Employed techniques like genomic selection to make genotype-phenotype predictions and GWAS to find gene(s) associated with a phenotype.
- Leveraged Nextflow pipelines, and parallel computing to improve workflow and manage big data.
- Led herbarium project discussions between lab manager, collaborators, and facilities. Communicated project goals.
- Conveyed research findings to researchers and lay audiences in meetings and conferences.

CDP North America, Disclosure Operations, New York, NY 11/20- 07/21
Disclosure Operations Intern (Remote)

- Analyzed disclosed supplier data, maintained database and provided major corporations and cities with visual displays of progress in emissions reduction (KPI metrics).
- Led testing for Scope 3 Report analysis.

- Collaborated with international team based in New York City and London.

Mount Holyoke College, Department of Institutional Research, South Hadley, MA 9/19-3/20
Data Analyst
- Conducted descriptive and quantitative analyses such as logistic regressions using R statistical software independently. Communicated findings to supervisors.
- Researched factors for retention in private universities and applied findings using variables from large survey-based database.

Harvard Forest, Petersham, MA 6/19-8/19
REU Researcher, Sydne Record, Ph.D., Paige Kouba, Ph.D. Student
- Created linear and logistic regression models to predict tree seedling growth rate and survival in R.
- Mentored student peer.

PUBLICATIONS

Vasquez Cerda, E.E, Bellis, E.S., Das, A., Slayton, E.R., Morris, G.P., & Lasky, J.R. (2026). Population genomic structure of sorghum landraces across landscape, environment and culture. *Molecular ecology*, e70287. https://doi.org/https://doi.org/10.1111/mec.70287 ***Invited for submission**

Lopez, L., Lang, P. L. M., Marciniak, S., Kistler, L., Latorre, S. M., Haile, A., **Vasquez Cerda, E**., Gamba, D., Xu, Y., Woods, P., Yifru, M., Kerby, J., McKay, J. K., Oakley, C. G., Ågren, J., Wondimu, T., Bulafu, C., Perry, G. H., Burbano, H. A., & Lasky, J. R. (2025). Museum Genomics Reveals Temporal Genetic Stasis and Global Genetic Diversity in Arabidopsis thaliana. *Molecular ecology*, e70081. https://doi.org/10.1111/mec.70081

HONORS & AWARDS

04/11/25 J. Ben and Helen D. Hill Memorial Fund
04/12/23 The Huck Institutes of the Life Sciences Graduate Travel Award
08/19/21 Homer F. Braddock Scholarship in Biology, Chemistry, and Physics
04/13/21 Eva J. Pell Distinguished Graduate Fellowship
03/23/21 **National Science Foundation GRFP**
02/25/21 Huck Institute Fellowship
08/2020 Princeton EEB Scholars Finalist

PRESENTATIONS

Applied Evolution Symposium, State College, PA 3/26
"The evolutionary processes underlying trichome dimorphism in natural *A. thaliana* populations"
Eberly Research Showcase: From Galaxies to Molecules, State College, PA 11/25
"Population genomic structure of sorghum landraces across landscape, environment and culture"
Ancient DNA Symposium, Welch Building, State College, PA 5/25
"Applications of historical DNA to unveil how genetic variation and microbiome diversity are maintained over space and time"

Plant Biology Research Showcase, Life Sciences Building, State College, PA 4/25
"Population genomic structure of sorghum landraces across landscape, environment and culture"
Eberly College of Science Benefactor's Day, LSB, State College, PA 3/25
"Applications of Herbarium Specimens in Evolutionary Biology"
Global Sorghum Conference, Montpellier, France 6/23
"Population genomic structure of sorghum landraces across landscape, environment and culture"
Mount Holyoke College, Earth Adventures, South Hadley, MA 9/19
"How many seedlings does it take to make a tree?"
Harvard Forest, 27th Annual Student Symposium, Petersham, MA 8/19
"How many seedlings does it take to make a tree?"

TEACHING AND OUTREACH

Biology 230 Lab: Molecules and Cells, State College, PA 1/26-Present
Teaching Assistant

- Taught 50 undergraduate students background and techniques in molecular biology such as gel electrophoresis, PCR, microscopy as well as bioinformatics skills such as BLAST on NCBI and Uniprot, MEGA software.
- Provided feedback to improve oral and writing skills.

Biology 220 Lab: Populations and Communities, State College, PA 8/25-12/25
Teaching Assistant

- Taught 50 undergraduate students ecological and evolutionary theory, statistical methods such as chi-squared tests, data-analysis on excel, hypothesis formulation and experimental setup.
- Provided feedback to improve oral and writing skills.
- Led students in ecological field research

BreakThrough New York, 123 William St. New York, NY 6/21- 5/25
Mentor

- Mentored undergraduate student in professional development, managing coursework, maintaining mental health and navigating student life.

DNA Day, Central Daulphin East HS, Harrisburg, PA 4/10/25
Penn. State Science Ambassador

- Taught forensics module and led activity for 32 students in an honors biology class and STEM career course.